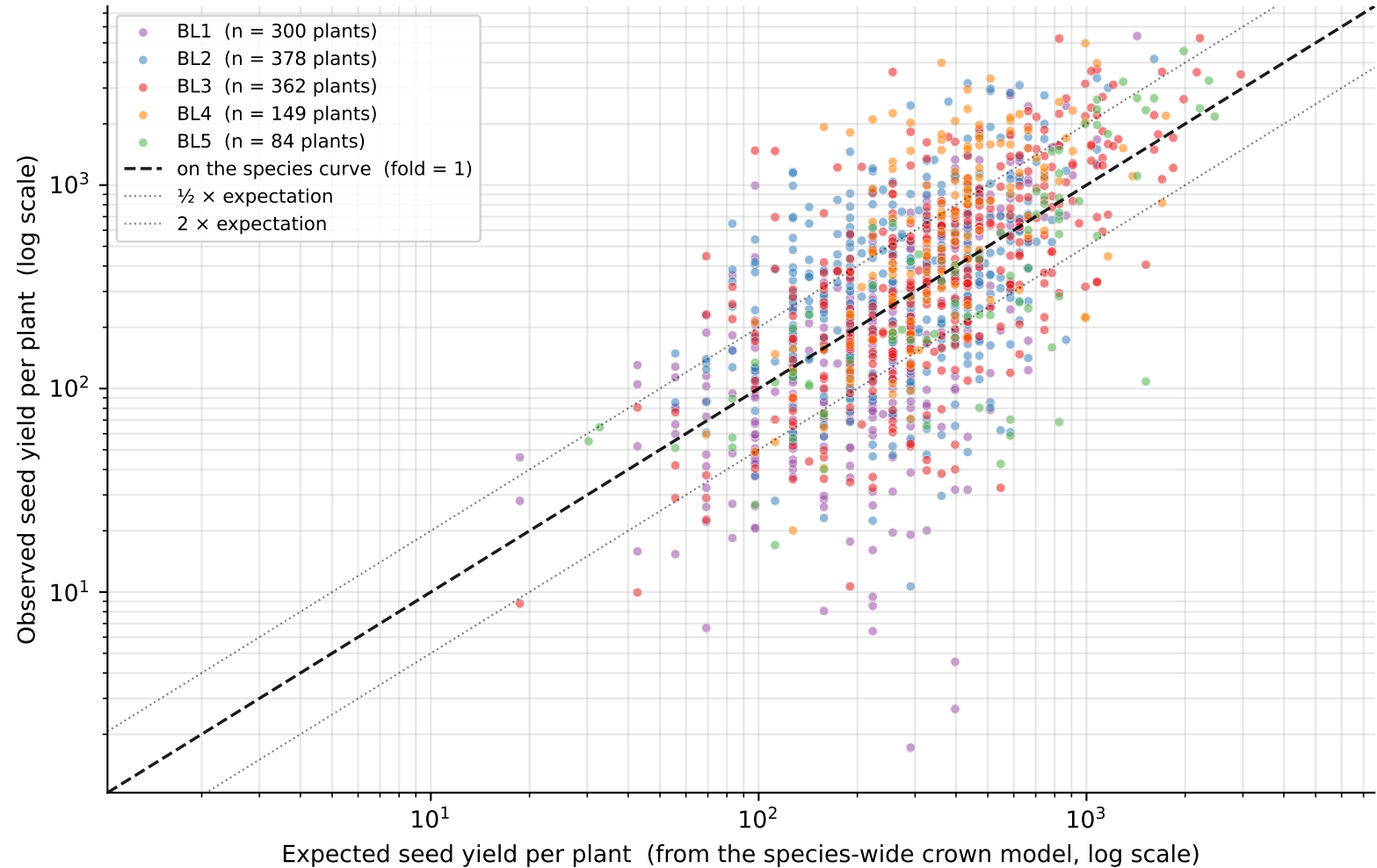


**A. Species-wide calibration scatter —  $R^2 = 0.37$  · one point per plant (n = 1273 plants, 37 populations, 2 years)** — the species-wide expectation is set



**Expectation model (STAGE 2)**

Winner (STAGE 1): **\*\*crown\*\***  
Formula:  $\log_{10}(\text{yield}) \sim \log c$   
 $R^2$  (in sample): 0.371  
Residual SD: 0.416  $\log_{10}$  ( $\approx \times / \div 2.6$ )  
Coefs: Intercept = +1.27,  $\log c$  = +1.19

**Predictor selection (STAGE 1, 10-fold CV)**

| predictor    | CV-RMSE | $\pm$ sd | $R^2$ | AIC    |
|--------------|---------|----------|-------|--------|
| crown        | 0.414   | 0.043    | 0.371 | 1379.6 |
| crown+height | 0.414   | 0.043    | 0.371 | 1381.1 |
| area         | 0.422   | 0.043    | 0.347 | 1428.6 |
| height       | 0.450   | 0.041    | 0.256 | 1593.5 |